

Graph-Based Compression of DEM-Resolved Agglomerate Morphology: A Preliminary Proof-of-Concept Study

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Abstract

The internal morphology of particle agglomerates can affect shielding and accessibility beyond conventional properties such as particle number, size and nominal fractal dimension. In this preliminary study, 135 synthetic monodisperse agglomerates were generated using a tunable particle-cluster/cluster-cluster method that enforces a prescribed mass-radius relation. The structures were relaxed using the discrete-element method (DEM) to test whether they remained stable under the selected particle and contact conditions. The resulting structures were converted into contact graphs and characterized using three graph-based shielding proxies: mean graph depth, shielded-particle fraction and accessibility score.

A geometry-aware graph neural network was evaluated using grouped five-fold cross-validation. The pooled graph representation was compressed into a latent bottleneck, and only this latent vector was supplied to the prediction head. A one-dimensional latent representation explained approximately 87 %, 89 % and 91 % of the variability in mean graph depth, shielded-particle fraction and accessibility score, respectively. These results demonstrate the feasibility of the DEM-to-graph-to-GNN workflow, although the current targets are geometric proxies rather than physical oxygen-transport quantities. The latent coordinate correlated strongly with particle number, radius of gyration, contact count and contact density. A separate interpretation analysis showed that conventional scalar descriptors could reconstruct approximately 99.6 % of its variation. The latent coordinate should therefore be interpreted as a compact size-extent-connectivity coordinate rather than an independent geometrical descriptor. Particle count alone reached a mean R^2 of at most 0.776 for predicting the three targets, compared with 0.892 for the GNN. Models using six conventional size and contact descriptors reached 0.891 with linear regression and 0.909 with a random forest. Thus, the GNN clearly improved

on particle count alone and achieved accuracy comparable to the multidescrptor models, while compressing the complete graph into a single learned coordinate.

Keywords: particle agglomerates; discrete-element method; contact graphs; graph neural networks; latent morphology representation; transport accessibility

1 Introduction

Transport models for biological and cohesive particle aggregates commonly describe morphology using a small set of integral quantities, such as aggregate diameter, porosity or fractal dimension. Such descriptors are useful, but they do not uniquely determine the internal arrangement of the primary particles. Aggregates with similar integral properties can still differ in connectivity, surface exposure, shielding and the existence of internal transport pathways. These structural differences may subsequently influence substrate or oxygen accessibility and therefore the effective activity of an aggregate.

The broader research concept is to derive morphology-aware oxygen-transport closures from particle-resolved CFD–DEM simulations and use them in coarse-scale CFD–PBM or Euler–Lagrange reactor models. In that future framework, a graph neural network would encode the detailed particle structure into a compact morphology descriptor, while a reduced closure would combine this descriptor with local reactor conditions to predict oxygen uptake, dynamic response and the fraction of oxygen-limited cells.

Before conducting expensive coupled flow and transport simulations, the present DEM-only study tests the feasibility of the structural part of this concept. Its central hypothesis is that a particle-contact graph contains nonlocal morphological information that can be compressed into a low-dimensional latent representation without losing the information required to predict internal shielding. The study therefore addresses three preliminary questions:

1. Can generated agglomerates be mechanically relaxed and converted consistently into particle-contact graphs?
2. Can a GNN predict nonlocal shielding and accessibility measures for parameter combinations withheld from training?
3. How many latent variables are required to retain this structural information?

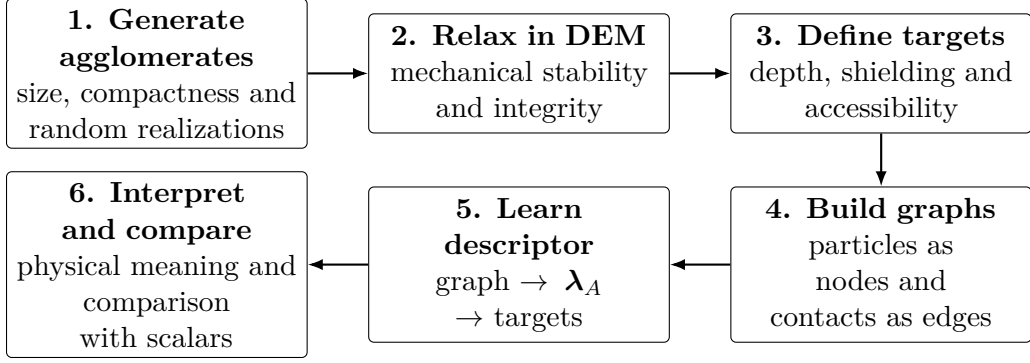


Figure 1. Six-step workflow used to generate and relax synthetic agglomerates, represent them as graphs, learn a compact structural descriptor and evaluate its physical meaning and predictive value.

The calculated targets are intentionally limited to geometric graph proxies. This separation is important: the study validates the computational workflow and the concept of morphology compression, while the physical relation to oxygen transport remains a task for the proposed particle-resolved project.

The proof-of-concept study proceeds in six steps. First, synthetic agglomerates are generated with different particle counts and physical sizes, as well as with equal particle counts but different compactness settings and random particle arrangements. Second, the artificial structures are relaxed in a physical DEM environment to test whether they remain mechanically stable under the selected particle and contact forces. Third, three structural targets are defined to quantify depth, shielding and accessibility relative to the outer surface. Fourth, each relaxed agglomerate is converted into a particle-contact graph. Fifth, a GNN is trained to compress the graph into one or more latent descriptors while retaining the information needed to predict the three targets. Finally, the learned descriptor is interpreted by comparing it with conventional physical and graph-based measures, and its predictive value is tested against particle count alone and against models using several conventional descriptors together. The complete workflow is summarized in Fig. 1.

2 Methodology and computational workflow

2.1 Synthetic agglomerate ensemble

The preliminary data set consists of monodisperse agglomerates generated with the tunable particle–cluster and cluster–cluster aggregation formulation of Filippov, Zurita and Rosner [1], following the efficient hierarchical implementation described by Skorupski et al. [2]. The method prescribes the mass–radius relation

$$N = k_f \left(\frac{R_g}{r_p} \right)^{D_f}, \quad (1)$$

where N is the number of primary particles, r_p is their radius, R_g is the radius of gyration including the finite primary-particle radius, D_f is the requested fractal dimension and k_f is the fractal prefactor.

Generation begins with stochastic 25-particle seed clusters. Starting from two touching particles, further particles are placed at random orientations on the radial shell required by Eq. (1); placements are accepted only when they establish contact without overlap. Equal-sized seed clusters are subsequently combined hierarchically. Before each merge, both subclusters are randomly rotated. Their prescribed center-of-mass separation R_{12} follows from the parallel-axis relation,

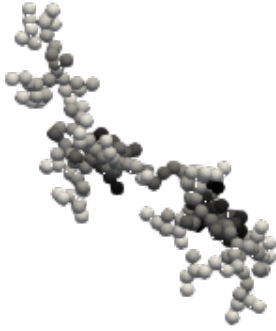
$$R_{12}^2 = r_p^2 \frac{N^2}{N_1 N_2} \left(\frac{N}{k_f} \right)^{2/D_f} - \frac{N}{N_2} R_{g,1}^2 - \frac{N}{N_1} R_{g,2}^2, \quad N = N_1 + N_2. \quad (2)$$

A Monte Carlo orientation search identifies a non-overlapping configuration whose first-contact separation agrees with R_{12} within the prescribed tolerance. The clusters are placed at this analytically determined first-contact distance, guaranteeing connectivity without overlap while allowing only a small, recorded deviation from the requested mass–radius law. The particle counts $N = 50, 100$ and 200 correspond to one, two and three hierarchical merging levels, respectively, from the common 25-particle seeds.

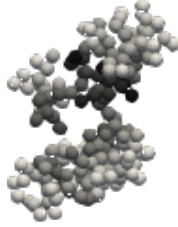
Three morphology settings were used: $(D_f, k_f) = (1.8, 1.3)$ for open, $(2.2, 1.1)$ for intermediate and $(2.6, 0.8)$ for comparatively compact agglomerates. The word “compact” is relative to the other fractal cases: the cluster–cluster construction creates contact-connected fractal structures and does not imply dense, nearly spherical packing. Dense

Table 1. Design variables used in the DEM investigation.

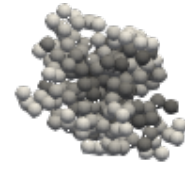
Design variable	Values
Primary-particle diameter d_p	1.0, 1.5, 2.0 μm
Number of particles N	50, 100, 200
Requested (D_f, k_f) and morphology	(1.8, 1.3) open; (2.2, 1.1) intermediate; (2.6, 0.8) comparatively compact
Random realizations	5 per parameter combination
Total number of cases	135



(a) Open aggregate



(b) Intermediate aggregate



(c) Comparatively compact aggregate

Figure 2. Representative post-DEM agglomerates illustrating the range of morphologies in the preliminary data set. The three agglomerates have the same primary-particle diameter ($d_p = 2 \mu\text{m}$) and particle count ($N = 200$) but different requested (D_f, k_f) morphology settings. The particles are coloured according to their graph depth h_i , as defined below, using a common colour scale ranging from 0 to 11.

spherical clusters would require an additional restructuring or densification procedure [3].

The data set comprises three primary-particle diameters, three particle counts, three morphology settings and five independent random realizations per design point,

$$3 \text{ diameters} \times 3 \text{ particle counts} \times 3 \text{ morphologies} \times 5 \text{ realizations} = 135 \text{ agglomerates.}$$

The details are summarized in Table 1.

Each realization was assigned a unique random seed. The generated particle coordinates and radii were exported in formats suitable for DEM relaxation, graph construction and visualization. All particles represented the same material; no permanent bonds were prescribed, and the initial translational and angular velocities were zero.

Table 2. Contact formulation used for DEM relaxation.

Model component	Physical formulation
Normal interaction	Hertzian elastic repulsion combined with regularized van der Waals attraction
Tangential interaction	History-dependent tangential force with Coulomb limitation
Rolling resistance	Directional rolling-resistance torque
Torsional resistance	Resistance to relative twisting at a contact
Near-contact regularization	Limitation of nonphysical normal-force states near contact
Permanent bonds	None

Table 3. Particle and contact parameters used in the DEM relaxation. The primary-particle diameter was varied, while the remaining properties were held constant.

Property	Symbol	Value
Particle shape	–	Sphere
Primary-particle diameter	d_p	1.0, 1.5, 2.0 μm
Reference particle density	ρ_p	1050 kg m^{-3}
Young’s modulus	E	$1.0 \times 10^9 \text{ Pa}$
Poisson ratio	ν	0.45
Coefficient of restitution	e	0.20
Sliding-friction coefficient	μ	0.30
Rolling-friction coefficient	μ_r	0.01
Hamaker constant	A_H	$3.028 \times 10^{-20} \text{ J}$
Minimum separation distance	$h_{\min}$	$4.0 \times 10^{-10} \text{ m}$

2.2 DEM relaxation and stability assessment

The generated structures were mechanically relaxed using a soft-sphere DEM formulation. Normal contact was described by Hertzian elastic repulsion and regularized van der Waals attraction. The adhesive interaction is governed by the Hamaker constant A_H and a minimum surface separation $h_{\min}$, which removes the singularity of the idealized interaction at vanishing separation [4, 5]. Tangential resistance was evaluated from the contact history and limited by Coulomb friction. Rolling and torsional resistance were also included. No permanent bonds were introduced; the relaxed structures were maintained solely by adhesion and contact friction. The applied contact models are summarized in Table 2.

Although the numerical setup permits multiple material types, identical properties were used for every interaction in the present monodisperse study.

The primary-particle diameters of 1.0–2.0 μm represent the characteristic micrometre scale of many bacterial cells [6]. The range is intentionally narrow: its purpose in this preliminary study is to test whether the graph workflow remains consistent when the physical length scale changes, while particle number and morphology are varied independently. The density $\rho_p = 1050 \text{ kg m}^{-3}$ represents a water-rich particle only slightly denser than the surrounding liquid. Measurements for *Escherichia coli*, for example, place the buoyant density near 1080–1105 kg m^{-3} [7, 8]. The slightly lower reference value used here is therefore an idealized near-neutral-buoyancy choice rather than a species-specific measurement. The Poisson ratio $\nu = 0.45$ represents a nearly incompressible effective particle.

The remaining mechanical parameters define a controlled relaxation model and must not be interpreted as a calibrated constitutive description of a particular microorganism. Reported effective bacterial-envelope moduli are commonly in the MPa range and depend strongly on species, physiological state and measurement protocol [9, 10]; the present value $E = 1 \text{ GPa}$ was selected as a stiff numerical contact parameter to restrict overlap during relaxation. The low restitution coefficient provides strong damping, whereas the sliding- and rolling-friction coefficients resist relative translation and rotation. The adopted Hamaker constant is of the conventional 10^{-20} J order associated with dispersion interactions, but its effective value depends on the interacting materials and the intervening liquid [5]. Accordingly, elasticity, dissipation, friction and adhesion should be varied in a sensitivity study and calibrated before applying the model to a specific bioaggregate. The relevant particle and contact parameters used in the present DEM relaxation simulations are listed in Table 3.

The relaxation simulations were performed with a physical time step of $\Delta t = 5.0 \times 10^{-11} \text{ s}$ and were run for up to 300000 time steps, corresponding to a maximum physical duration of $1.5 \times 10^{-5} \text{ s} = 15 \mu\text{s}$. The purpose of this DEM stage was to obtain mechanically stable structures and to verify that the artificially generated agglomerates remained intact under the selected elasticity, dissipation, friction and adhesive-contact conditions without permanent bonds. The suitability of the time step was checked every 1000 steps against the characteristic contact and neighbour-list time scales. Particle and pair-interaction data were written every 20000 steps, corresponding to output intervals of $1 \mu\text{s}$. Gravity was not activated because the simulation focused on internal contact-scale relaxation

without sedimentation or an externally imposed loading direction.

The particle output contained positions, radii, particle IDs and velocities, whereas the pair output contained interacting particle IDs, gap, overlap, contact area, normal and tangential forces, total force and torque. These data were converted to VTK for inspection and subsequent graph construction. Because the adhesive interaction also acts across a finite gap, a reported particle pair does not necessarily represent a geometrically touching contact. The pairs were therefore classified using

$$\delta_{ij} = r_i + r_j - \|\mathbf{x}_i - \mathbf{x}_j\|. \quad (3)$$

Pairs satisfying $\delta_{ij}/d_{\text{ref}} \geq -10^{-6}$ were treated as touching. This small tolerance retains numerically near-touching pairs while excluding separated neighbours interacting only through finite-range adhesive forces.

For each case, particle and contact VTK files were matched by time step. An agglomerate was considered relaxed when, over three consecutive output intervals,

$$\frac{\Delta x_{\text{RMS}}}{d_p} \leq 10^{-4}, \quad 1 - \frac{|C_k \cap C_{k+1}|}{|C_k \cup C_{k+1}|} \leq 0.01, \quad (4)$$

where C_k denotes the set of particle contacts at time step k . These criteria require both the particle positions and the contact network to become nearly stationary.

Agglomerate integrity was evaluated independently using connected-component analysis. Each particle was treated as a graph node and every reported particle-pair interaction as an undirected edge. All 135 final structures contained exactly one connected component with a largest-component fraction of unity. Thus, none of the artificially generated agglomerates fragmented during the DEM relaxation.

2.3 Effective fractal-dimension characterization

The fractal dimensions specified during agglomerate generation define the requested morphology settings. Since the generated structures contain only 50–200 primary particles and are subsequently relaxed by DEM, these input values should not be interpreted as exact measured properties of every final structure. The relaxed agglomerates were therefore characterized using two complementary finite-scale methods. Both methods are commonly used to characterize particle aggregates, although their estimates can depend

on the available scale range and the finite number of primary particles [11, 12].

The first estimate was based on the mass–radius scaling relation commonly used for fractal particle aggregates [13, 11],

$$N(r) \propto r^{D_f}, \quad (5)$$

where $N(r)$ is the number of particle centres within a distance r from the agglomerate centre of mass. After calculating the distance of every particle centre from the agglomerate centre of mass, the particles were sorted according to this radial distance. The effective fractal dimension was then obtained from the slope of

$$\log N(r) = D_f \log r + C. \quad (6)$$

Only the range containing 10–80% of the particles was included in the regression. The innermost region was excluded because it contains only a small number of particles, while the outermost region is strongly affected by the finite boundary of the agglomerate.

A second estimate was obtained using the box-counting method [14, 15, 12]. The particle centres were covered by a three-dimensional Cartesian grid with box size ϵ , and the number of boxes containing at least one particle centre, $N_{\text{box}}(\epsilon)$, was counted. The effective box-counting dimension follows from

$$N_{\text{box}}(\epsilon) \propto \epsilon^{-D_f}, \quad (7)$$

and was calculated as the slope of $\log N_{\text{box}}$ against $\log(1/\epsilon)$.

Before box counting, each agglomerate was translated to its centre and aligned with its principal axes. Twelve logarithmically spaced box sizes were evaluated. For each box size, 32 shifted grids were tested and the smallest occupied-box count was retained to reduce sensitivity to the arbitrary grid origin. Scales for which nearly all particles occupied separate boxes, as well as scales for which the complete agglomerate occupied approximately one box, were excluded. Such scale selection is important because both limits contain little information about the intermediate spatial scaling [15, 12].

2.4 Graph representation and structural targets

Each selected agglomerate snapshot was represented as a contact graph. Primary particles form the nodes, while only geometrically touching or numerically near-touching pairs satisfying $\delta_{ij}/d_{\text{ref}} \geq -10^{-6}$ form the edges. Separated attractive neighbours were excluded. This choice gives the graph a clear geometrical interpretation and prevents its topology from depending on the finite interaction cutoff of the adhesive model. Every retained pair was stored in both directions for message passing. Thus, an agglomerate with N particles and N_c retained contacts has N nodes and $2N_c$ directed edges. All 135 agglomerates yielded valid graphs and were included in the analysis; no case was skipped.

Three node features were available after preprocessing. The minimal model used the two nonconstant structural scalars

$$\mathbf{x}_i = \left[\frac{|\mathbf{x}_i - \mathbf{x}_c|}{R_g}, z_i \right], \quad (8)$$

where d_{ref} is twice the median particle radius, $\mathbf{x}_c$ is the agglomerate center, R_g is the radius of gyration, and z_i is the coordination number. The third available node feature, r_i/d_{ref} , is approximately 0.5 for all present monodisperse particles and was therefore omitted from the minimal model.

The processed DEM data contained 22 available contact features, including normalized distance and overlap, gap, contact area, force and torque quantities and contact-state flags. For the minimal model, only the normalized centre distance d_{ij}/d_{ref} was selected from these stored contact features, because connectivity and interparticle geometry are the quantities directly relevant to the three structural targets considered here. The graph connectivity and normalized relative-position vector were additionally used for message passing. The remaining available contact features were retained for possible sensitivity studies but were not supplied to the reported model.

The convex hull of the particle centres was calculated using the Qhull option QJ. Particles located at the hull vertices were classified as surface particles and assigned a graph depth of $h_i = 0$. The graph depth of every other particle was defined as the minimum number of contact connections required to reach a surface particle.

Three aggregate-level quantities were then calculated:

$$\bar{h} = \frac{1}{N} \sum_{i=1}^N h_i, \quad (9)$$

$$f_{h \geq 2} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(h_i \geq 2), \quad (10)$$

$$S_{\text{access}} = \frac{1}{N} \sum_{i=1}^N \exp(-\beta h_i), \quad \beta = 0.5. \quad (11)$$

The mean graph depth $\bar{h}$ indicates how deeply the particles are located within the aggregate. A larger value represents a more strongly shielded internal structure. The quantity $f_{h \geq 2}$ is the fraction of particles that are separated from the surface by at least two contact connections and therefore represents the strongly shielded particle fraction. Finally, S_{access} is a continuous accessibility score: surface particles contribute 1, while the contribution of internal particles decreases exponentially with graph depth. Consequently, values closer to one indicate a more accessible aggregate, whereas lower values indicate stronger internal shielding.

These quantities are deliberately defined geometric proxies derived from the contact network. They test whether a GNN can distinguish aggregate structures with different degrees of internal shielding; they do not yet represent oxygen transport or oxygen uptake.

2.5 GNN compression and training

In addition to the stored node and edge features, the model uses the particle coordinates relative to the aggregate centre and normalized by the reference diameter d_{ref} . The corresponding relative-position vector is assigned to every directed edge. Random rigid rotations are applied during training to prevent the model from associating the target quantities with a particular aggregate orientation.

The node and edge inputs are embedded into a hidden space of dimension 64. Six residual GINE message-passing layers with layer normalization and dropout are then applied. The message-passing depth allows information to propagate over extended paths through the aggregate, while the residual connections reduce optimization difficulties and the loss of local information. Graph-level mean, maximum and size-sensitive normalized-sum pooling are combined to form a graph-derived aggregate representation. This

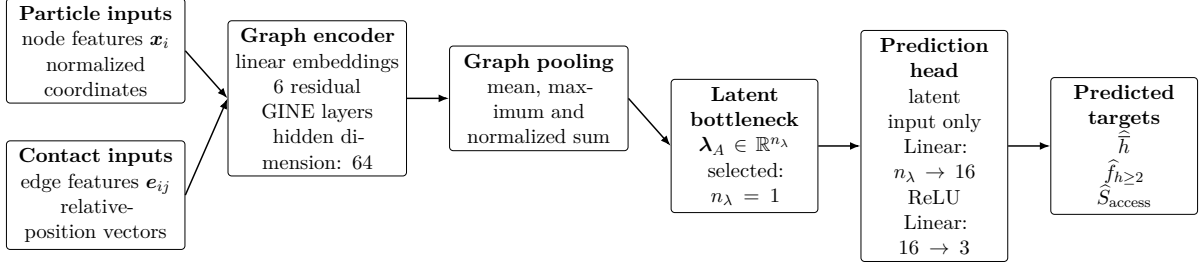


Figure 3. Architecture of the graph-based morphology model. Particle and contact information is processed by the message-passing graph encoder. The pooled graph representation is compressed into the latent morphology vector λ_A . A two-layer multilayer perceptron maps only this latent vector to the three supervised geometric targets. All components are trained jointly.

representation alone is compressed through the latent bottleneck into a morphology vector

$$\lambda_A = [\lambda_1, \dots, \lambda_{n_\lambda}] \in \mathbb{R}^{n_\lambda}, \quad (12)$$

where the latent dimension n_λ is treated as a model hyperparameter. Candidate values $n_\lambda \in \{1, 2, 3, 4\}$ are evaluated by grouped cross-validation. Thus, the GNN learns the values of the latent components, whereas their required number is determined from the predictive performance and model-complexity comparison.

No separately calculated aggregate-level descriptors are supplied to either the latent bottleneck or the prediction head. Aggregate size, spatial extent and connectivity can nevertheless be inferred from the number and arrangement of nodes, their normalized coordinates, the contact topology and the size-sensitive normalized-sum pooling term. A two-layer prediction head maps only λ_A to the three supervised proxy targets,

$$\hat{\mathbf{y}}_A = \left[\hat{h}, \hat{f}_{h \geq 2}, \hat{S}_{\text{access}} \right]. \quad (13)$$

Training used Adam with a learning rate of 10^{-3} , weight decay of 10^{-5} , batch size 16 and random seed 7. The targets were standardized using only the respective training data. A validation subset and early stopping were used to retain the best model in each fold instead of automatically selecting the last training epoch.

The assessment used grouped five-fold cross-validation. All five random realizations with the same (d_p, N, D_f, k_f) combination were assigned to the same test fold. This produced 27 parameter groups. Consequently, each reported prediction was produced by a model that had not seen another realization of the corresponding parameter combination

Table 4. Influence of the latent dimension on the combined out-of-fold coefficient of determination. The selected configuration is highlighted in bold.

n_λ	R_h^2	$R_{f_{h \geq 2}}^2$	$R_{S_{\text{access}}}^2$	Mean R^2
1	0.869	0.894	0.913	0.892
2	0.878	0.880	0.902	0.887
3	0.867	0.886	0.912	0.888
4	0.869	0.870	0.889	0.876

during training. The predictions from the five held-out folds were combined to calculate the out-of-fold metrics

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{j=1}^n (y_j - \hat{y}_j)^2}, \quad (14)$$

$$\text{MAE} = \frac{1}{n} \sum_{j=1}^n |y_j - \hat{y}_j|, \quad (15)$$

$$R^2 = 1 - \frac{\sum_{j=1}^n (y_j - \hat{y}_j)^2}{\sum_{j=1}^n (y_j - \bar{y})^2}. \quad (16)$$

Here, RMSE and MAE quantify the prediction error, whereas R^2 measures the fraction of the variation among the 135 aggregates explained by the model.

3 Results and interpretation

3.1 Latent dimension and predictive performance

Selection of the latent dimension. Latent dimensions $n_\lambda = 1, 2, 3$ and 4 were evaluated using the same grouped cross-validation procedure. Table 4 summarizes the combined out-of-fold R^2 values.

The one-dimensional representation reached a mean $R^2 = 0.892$. Although the individual best target-specific values were distributed among several latent dimensions, the one-dimensional model had the highest mean performance. It is therefore selected both by predictive performance and by the predefined compactness rule, which selects the smallest latent dimension whose mean R^2 lies within 0.02 of the maximum. Accordingly, $n_\lambda = 1$ was selected. The final latent morphology representation is therefore

$$\boldsymbol{\lambda}_A = [\lambda_1]. \quad (17)$$

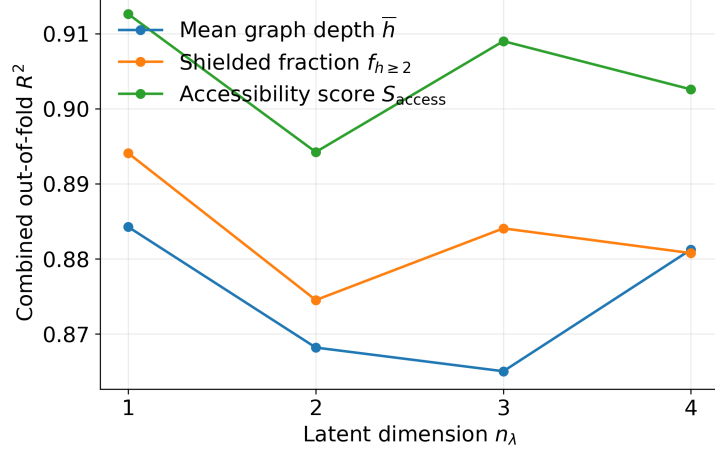


Figure 4. Selection of the latent morphology dimension based on grouped five-fold cross-validation.

Table 5. Out-of-fold performance of the selected GNN with one latent morphology descriptor.

Target	RMSE	MAE	R^2
$\bar{h}$	0.390	0.293	0.869
$f_{h \geq 2}$	0.049	0.039	0.894
S_{access}	0.038	0.030	0.913

It provides both the most compact representation and the highest predictive performance among the tested latent dimensions. Because the same grouped cross-validation predictions were used both to compare candidate dimensions and to select the final dimension, these scores are model-development estimates. The selection should be confirmed using new realizations, repeated grouped cross-validation or nested cross-validation.

Performance of the selected model. For the selected one-dimensional latent representation, the combined out-of-fold metrics were:

The selected model explains approximately 87% of the variation in mean graph depth, 89% of the variation in the shielded-particle fraction and 91% of the variation in the accessibility score. These results provide a successful proof of concept for the complete DEM-to-graph-to-GNN workflow under the more restrictive grouped evaluation. Nevertheless, the targets remain deliberately defined geometric proxies rather than oxygen-transport quantities, and the data set contains 135 structures. Additional independent realizations and subsequent validation against resolved transport calculations are therefore required.

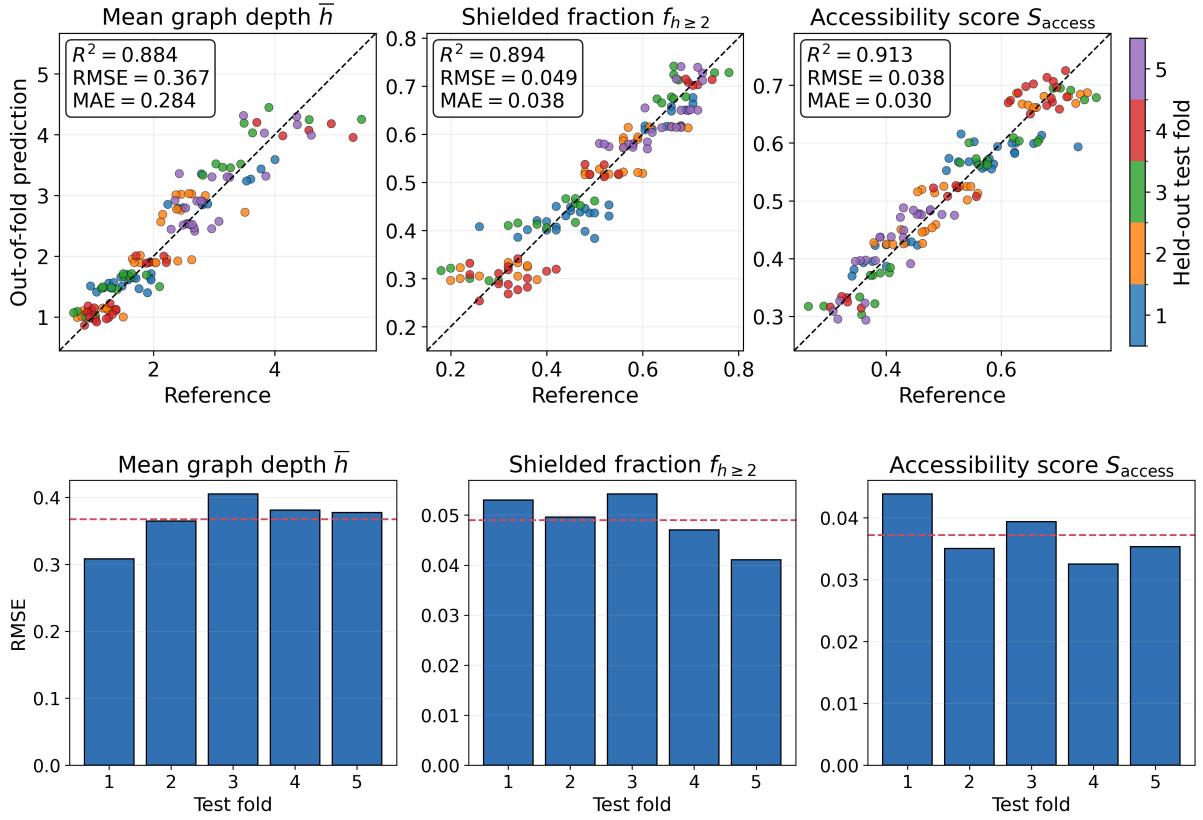


Figure 5. Performance of the geometry-aware GNN under grouped five-fold cross-validation: (top) combined out-of-fold parity plots and (bottom) variation of the RMSE among the five held-out folds.

3.2 Physical interpretation of the learned descriptor

The latent coordinates obtained from independently trained cross-validation models cannot be compared directly because their sign, scale and offset are not identifiable across separate training runs. The descriptor analysis therefore did not combine the fold-specific latent values generated during cross-validation. Instead, after selecting $n_\lambda = 1$, final models were trained on all 135 graphs using three random seeds (7, 17 and 27). The resulting one-dimensional coordinates were standardized, aligned in sign and averaged to obtain a consistent descriptor λ_1 for descriptive interpretation. This all-data representation was used only to examine the information encoded by the network; it was not treated as a leakage-free test feature.

For comparison with the learned coordinate, a set of conventional scalar agglomerate and contact-network descriptors was considered. It comprised the particle count N , primary-particle diameter d_p , requested fractal dimension D_f , fractal prefactor k_f , dimensionless radius of gyration R_g/d_{ref} , number of retained contacts N_c , mean and maximum coordination numbers $\bar{z}$ and z_{max} , and contact density

$$\rho_c = \frac{2N_c}{N(N-1)}. \quad (18)$$

The contact density represents the fraction of all possible particle pairs that are connected. These scalar descriptors summarize aggregate size, spatial extent and contact connectivity without retaining the complete particle-contact graph. In the present analysis, D_f denotes the requested generation value rather than a measured post-relaxation fractal dimension.

Pearson and Spearman correlations were calculated between λ_1 and these conventional scalar descriptors. Pearson's r measures linear association, whereas Spearman's ρ measures monotonic association based on the ranked values. Table 6 summarizes the results.

Particle count showed the strongest monotonic association with the latent coordinate, with $r = 0.920$ and $\rho = 0.943$. Radius of gyration was similarly strongly associated ($r = 0.922$, $\rho = 0.921$). The number of contacts also showed a strong positive correlation because it naturally increases with aggregate size, approximately as

$$N_c \simeq \frac{N\bar{z}}{2}. \quad (19)$$

Table 6. Association between the learned latent coordinate λ_1 and conventional scalar agglomerate and contact-network descriptors. The descriptors are ordered by the absolute Spearman correlation.

Descriptor	Pearson r	Spearman ρ
Particle count N	0.920	0.943
Contact density ρ_c	-0.878	-0.925
Radius of gyration R_g/d_{ref}	0.922	0.921
Number of contacts N_c	0.918	0.876
Mean coordination $\bar{z}$	0.435	0.470
Requested fractal dimension D_f	-0.346	-0.313
Fractal prefactor k_f	0.336	0.313
Maximum coordination z_{max}	0.087	0.130
Primary-particle diameter d_p	0.004	0.002

Conversely, contact density decreases with N because the number of possible particle pairs increases quadratically. The correlations with particle count, contact count and contact density should therefore not be interpreted as three independent effects; they primarily reflect the same dominant size dependence.

The strong association with radius of gyration and the weaker association with mean coordination indicate that the coordinate combines aggregate size, spatial extent and connectivity. Requested D_f and k_f showed modest and oppositely signed associations. In the present design these parameters are paired one-to-one across the three morphology classes, so their separate effects cannot be identified. Primary-particle diameter showed essentially no association, consistent with normalization by d_{ref} . A measured post-relaxation fractal dimension and porosity should be included in a subsequent analysis.

The five realizations within each of the 27 parameter groups produced a mean within-group standard deviation of 0.050 in the aligned, standardized latent coordinate; the group-wise values ranged from 0.015 to 0.147. Thus, the learned coordinate varies only moderately among repeated stochastic realizations compared with its overall unit standard deviation.

To examine whether the latent coordinate contains information beyond these scalar descriptors, supplementary regression models were trained to reconstruct λ_1 . Here, reconstruction means predicting the latent value assigned to an agglomerate from either an individual descriptor or a combination of descriptors, without using the complete particle-contact graph as an input.

The term “combined linear” denotes one multivariable ordinary linear regression using

Table 7. Grouped cross-validation performance for reconstructing the learned latent coordinate from conventional scalar agglomerate and contact-network descriptors.

Reconstruction model	Grouped-CV R^2
Combined linear model	0.996
Combined random forest	0.996
Linear model: R_g/d_{ref}	0.843
Linear model: particle count	0.827
Linear model: number of contacts	0.823
Linear model: contact density	0.758
Linear model: mean coordination	0.164
Linear model: requested D_f	-0.046
Linear model: fractal prefactor k_f	-0.048
Linear model: maximum coordination	-0.067
Linear model: particle diameter	-0.121

all nine numeric descriptors listed above simultaneously, together with a one-hot encoding of the morphology label. It is therefore a combination of conventional descriptor inputs, not a combination or ensemble of GNN models. The “combined random forest” uses the same descriptor set but allows nonlinear relations and interactions. Because morphology is assigned directly from the paired (D_f, k_f) settings in this data set, that categorical input is partly redundant with those two numeric variables.

The reconstruction models were evaluated using grouped five-fold cross-validation. All five random realizations belonging to the same (d_p, N, D_f, k_f) parameter combination were retained in the same fold. During each iteration, a regression model was fitted using the remaining four folds and applied to the withheld parameter combinations. This grouping prevents realizations of the same design point from appearing in both the training and test data. The predictions from all five held-out folds were then combined to calculate the out-of-fold coefficient of determination reported in Table 7.

A linear model using R_g/d_{ref} alone reconstructed 84.3% of the grouped out-of-fold variation in λ_1 , while particle count and contact count alone reconstructed 82.7% and 82.3%, respectively. Combining the conventional descriptors increased R^2 to approximately 99.6% for linear regression ($R^2 = 0.996031$) and a shallow random forest ($R^2 = 0.995844$). The negligible difference between the combined linear and nonlinear models indicates that the learned coordinate is almost a linear combination of the available conventional descriptors over the investigated design space. The negative R^2 values for several individual design variables indicate that they alone generalize worse than predicting the mean latent

value for held-out parameter groups.

This reconstruction analysis serves to interpret the fixed latent representation rather than to provide an independent validation of the complete GNN workflow. The latent values were obtained from final GNNs trained on the complete data set, whereas grouped cross-validation was applied only to the subsequent descriptor-to-latent regression models. A fully independent assessment would require the latent representation and its reconstruction model to be fitted within a nested cross-validation procedure.

Consequently, the selected latent coordinate should not presently be regarded as a new independent geometrical descriptor. It is more accurately interpreted as a compact size–extent–connectivity coordinate learned by the end-to-end model. Particle number is its strongest monotonic correlate, but the reconstruction results show that radius of gyration and contact-network descriptors also contribute materially.

The graph-derived latent coordinate remains strongly size-dependent because particle number is implicit in the number and topology of graph nodes, while the normalized-sum pooling term is also size-sensitive. The supervised targets vary systematically with aggregate size. A latent coordinate learned from the graph should therefore not automatically be interpreted as size-independent. Future ablation studies should compare mean–maximum pooling with and without the normalized-sum term and assess prediction within fixed- N subsets. The following comparison with conventional scalar descriptors tests how much predictive value the present graph model adds beyond aggregate-level summaries.

3.3 Comparison with conventional scalar descriptors

The preceding analysis showed that the learned coordinate is strongly related to particle count: N had a Spearman correlation of 0.943 with λ_1 , and a linear model based on N alone reconstructed approximately 83 % of the variation in λ_1 . At the same time, the complete GNN model, in which the three targets are predicted only through λ_1 , explained about 90 % of their variation on average. This raises an important practical question: if the learned coordinate is so strongly related to N , could the GNN be omitted and particle count used directly as the only structural descriptor, with similar or even better predictive accuracy?

To answer this question, four simple models were trained using only N as input: linear

Table 8. Predictive performance of the GNN, particle-count-only models and models using six conventional descriptors. Higher R^2 indicates that a larger fraction of the variation in the target is explained.

Model	R_h^2	$R_{fh \geq 2}^2$	$R_{S_{\text{access}}}^2$	Mean R^2
Combined-descriptor random forest	0.894	0.905	0.926	0.909
GNN	0.869	0.894	0.913	0.892
Combined-descriptor linear model	0.894	0.870	0.909	0.891
Count-only random forest	0.666	0.831	0.832	0.776
Count-only quadratic	0.666	0.830	0.832	0.776
Count-only categorical	0.666	0.830	0.832	0.776
Count-only linear	0.677	0.785	0.800	0.754

regression, quadratic regression, a categorical model that treats $N = 50, 100$ and 200 as three separate size classes, and a shallow random forest. These alternatives range from a simple linear relation to more flexible relations between particle count and the targets.

A stricter comparison was then performed using six conventional descriptors calculated from each relaxed agglomerate and its contact graph:

$$N, \quad \frac{R_g}{d_{\text{ref}}}, \quad N_c, \quad \rho_c, \quad \bar{z}, \quad z_{\text{max}}. \quad (20)$$

These quantities describe particle number, spatial extent and contact connectivity without retaining the complete graph. They were supplied together to a multivariable linear model and to a shallow random forest. Requested D_f , k_f and the morphology-class label were excluded because they are generation settings rather than independently calculated properties of the relaxed structure.

All baseline models were tested using the same five data splits as the GNN. In every split, all five realizations of a given (d_p, N, D_f, k_f) combination were kept together and used either for training or for testing. The comparison is therefore based on exactly the same unseen agglomerate groups for every model.

Particle count alone already described a substantial part of the variation in the three targets. However, none of the count-only models matched the GNN. The best count-only result had a mean R^2 of 0.776 , whereas the GNN reached 0.892 . Relative to the strongest count-only result for each target, the GNN increased R^2 by approximately 0.192 for mean graph depth, 0.063 for the shielded-particle fraction and 0.081 for accessibility. The largest

improvement was obtained for mean graph depth, which depends not only on how many particles are present, but also on how they are arranged and connected to the aggregate surface.

The quadratic, categorical and random-forest models produced almost identical results. The lower performance of the count-only models is therefore not simply caused by assuming a linear relationship with N . Particle count is an important descriptor of aggregate size, but it cannot distinguish agglomerates with the same N and different particle arrangements.

The comparison with the six-descriptor models gives a more qualified result. The combined linear model achieved a mean R^2 of 0.891, close to the GNN value of 0.892. The combined random forest reached 0.909 and therefore exceeded the GNN by 0.016 in mean R^2 . For the present regular synthetic ensemble and graph-derived targets, detailed graph encoding is consequently not required to obtain the highest predictive accuracy. A small set of carefully chosen scalar descriptors contains nearly all information needed for these three quantities.

There are nevertheless reasons to retain the GNN route for the planned study. First, it compresses the complete graph into one learned coordinate, whereas the conventional models require six separately selected and calculated inputs. Second, the representation is learned directly for the prediction task rather than fixed in advance, which makes it adaptable when polydispersity, irregular shapes, branching, restructuring or additional contact information are introduced. Third, the present targets are themselves geometric summaries of the same contact graph and are therefore expected to be well represented by simple graph-derived scalars. Physically resolved transport targets may depend on spatial pathways and local arrangements that these scalars do not retain. The current result therefore supports the GNN as a compact and extensible representation, but it does not yet demonstrate that a GNN is necessary. That claim must be tested on a broader morphology ensemble and, most importantly, against independently calculated diffusion and oxygen-transport quantities.

4 Conclusions and next steps

This preliminary investigation establishes a complete and reproducible path from generated particle agglomerates through DEM relaxation and graph construction to a compact

learned morphology representation. The DEM stage also fulfilled its intended stability-test role: all 135 synthetic agglomerates remained single connected structures under the selected particle and adhesive-contact conditions. The grouped cross-validation results show that the GNN does not merely reproduce individual training realizations: it retains predictive capability for withheld (d_p, N, D_f, k_f) combinations. The selection of one latent component is computationally attractive for the planned multiscale framework. Moreover, the GNN increased the mean R^2 from approximately 0.776 for the best particle-count-only baseline to 0.892, demonstrating that the complete graph provides useful predictive information beyond aggregate size alone. A linear model using six conventional descriptors achieved a comparable mean R^2 of 0.891, while a random forest using the same descriptors reached 0.909 and slightly outperformed the GNN. The current targets therefore do not establish a predictive necessity for graph encoding. The learned coordinate is nevertheless useful as a compact representation: it combines information distributed across the particle-contact graph into one coordinate that supports accurate prediction of all three targets, rather than requiring six separately selected scalar inputs. Its interpretation showed that λ_1 is an effective compressed size–extent–connectivity coordinate for the selected shielding proxies, not a newly discovered independent geometrical quantity.

The present evidence should nevertheless be interpreted as a methodological proof of concept. The three supervised targets are constructed from the convex hull and contact graph and are not independent measurements of mass transfer. Furthermore, convex-hull vertices provide only a simplified description of surface accessibility and do not resolve concave surfaces, pores or transport channels. The current set of 135 structures still limits the statistical assessment.

The close performance of the GNN and the combined-descriptor models must be viewed in relation to the deliberately structured data set. Even within the present design, agglomerates of comparable overall extent can arise from three primary-particle diameters, despite their relatively narrow range, three fractal-dimension or compactness settings, and five stochastic realizations of each parameter combination. Thus, the same particle count does not define a unique morphology. Natural agglomerates are expected to display substantially greater diversity in particle-size distribution, shape, porosity, branching, local packing and contact connectivity, even when their overall dimensions are similar. As this morphological diversity is introduced into a larger data set, particle count and other

low-order scalar measures may become less complete descriptions of structure. A learned graph representation could then become more valuable for distinguishing agglomerates with similar standard descriptors but different internal organization.

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